```
#fibonacci

.data
ent: .asciiz "Please enter how many fibonacci numbers you want:"
message: .asciiz " number of fibonacci series is stored at memory location "
series: .asciiz "fibonacci series: "

.text
main:
li $v0, 4
la $a0, ent
syscall

li $v0, 5
syscall

move $s0, $v0 #here we have stored n.
ble $s0, $0, end #if user input is non-positive print nothing

#allocating space to save
#li $t0, 4
#mul $a0, $t0, $s0
sll $a0, $s0, 2
addi $a0, $a0, 8
li $v0, 9
syscall
move $s7, $v0 #base address of the number store.
move $s6, $s7 #the memory pointer we will advance



#initialising with first two numbers
li $s1, 0
sw $s1, 0($s6)
li $s2, 1
#addi $s6, $s6, 4
sw $s2, 4($s6)
li $s3, 0 #counter


li $v0, 1
addi $a0, $s3, 1
syscall
li $v0, 4
la $a0, message
syscall
li $v0, 1
move $a0, $s6
syscall
li $v0, 11
li $a0, 10 #newline char
syscall

addi $s3, $s3, 1
addi $s6, $s6, 8 #this command here so 0 fibonacci printed correctly.

loop:
beq $s3, $s0, boop

#print the new number
li $v0, 1
addi $t1, $s3, 1 #counter starts from 0.
move $a0, $t1
syscall
li $v0, 4
la $a0, message
syscall
```

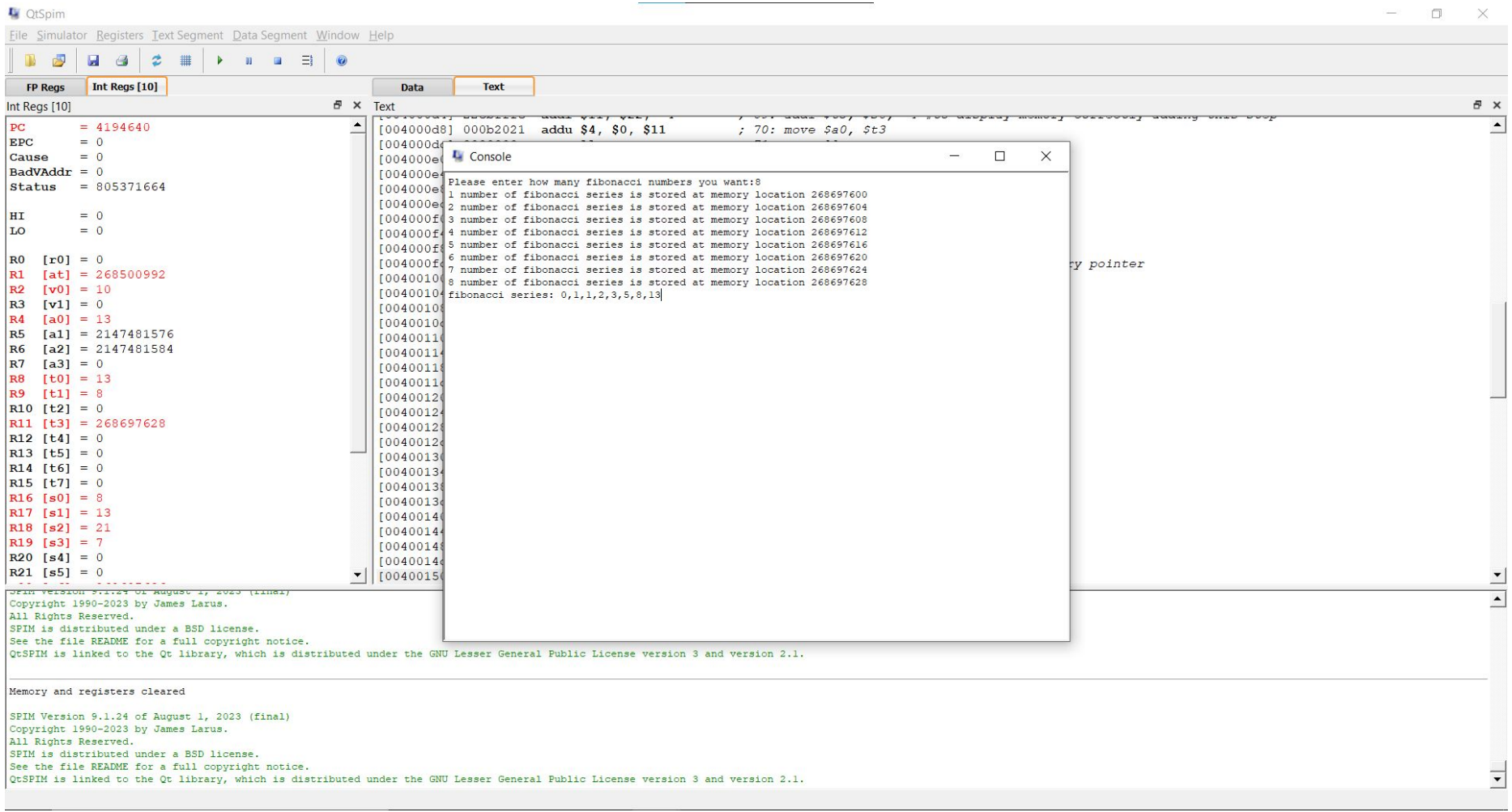
```
li $v0, 1
addi $t3, $s6, -4 #to display memory correctly adding this step
move $a0, $t3
syscall
li $v0, 11
li $a0, 10 #newline char
syscall

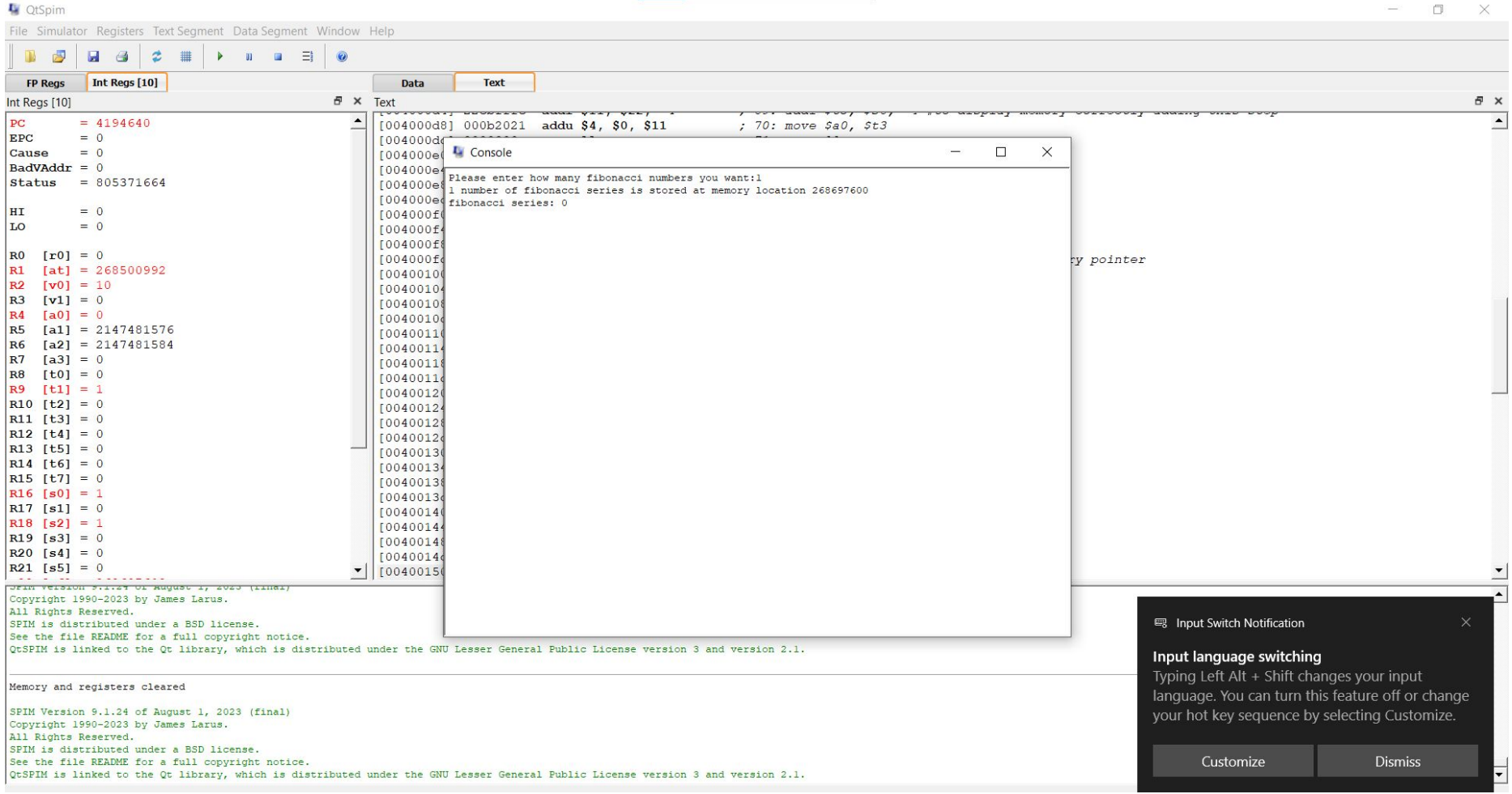
move $t0, $s2
add $s2, $s2, $s1
move $s1, $t0
sw $s2, 0($s6)
addi $s6, $s6, 4 #advancing the memory pointer
addi $s3, $s3, 1
j loop

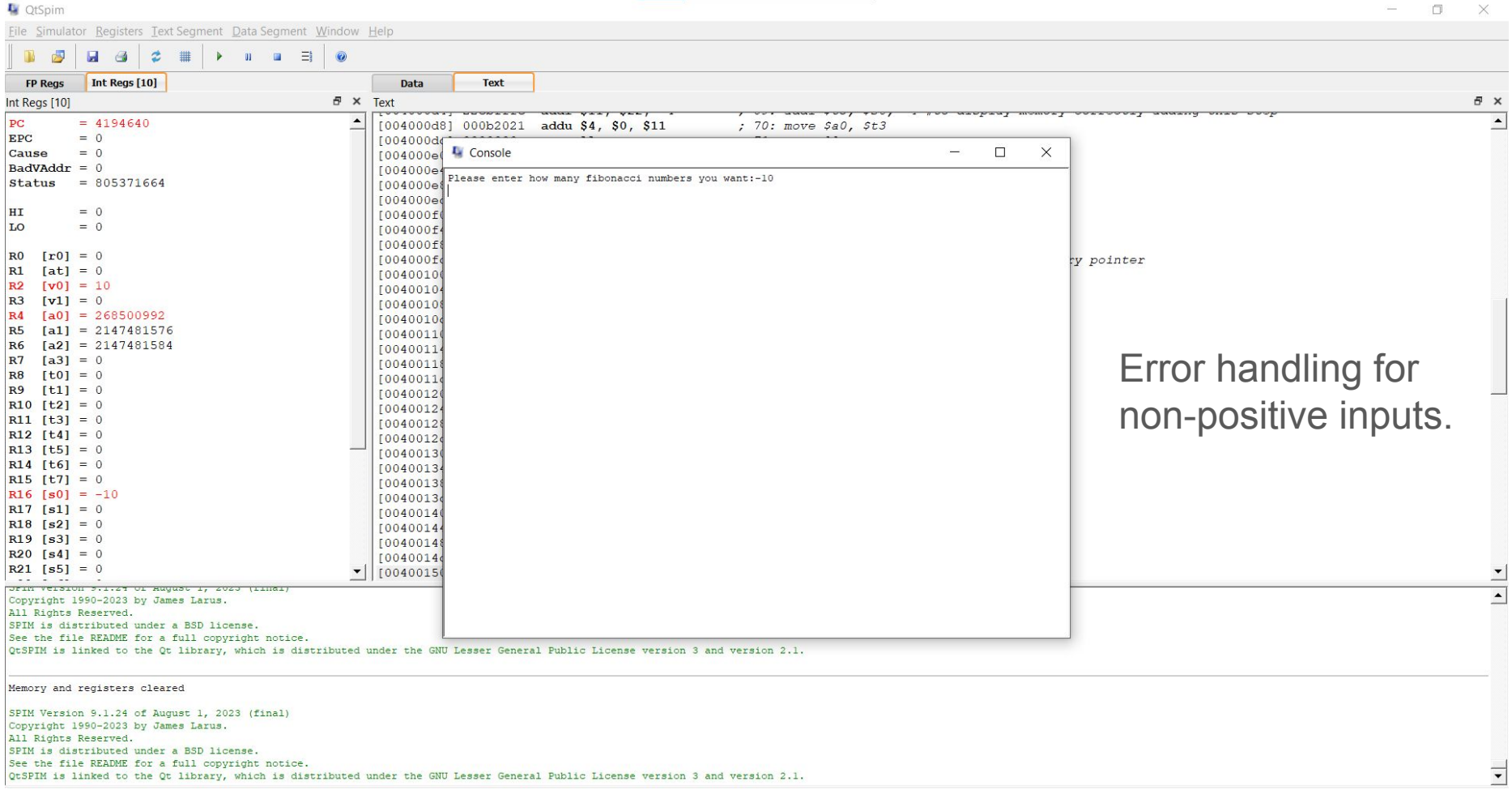
#now printing the final series of numbers
boop:
li $v0, 4
la $a0, series
syscall
li $s3, 0
goop:
beq $s3, $s0, end
lw $a0, 0($s7)
li $v0, 1
syscall
#special beq check to prevent the last comma
addi $t1, $s3, 1
beq $t1, $s0, end
li $v0, 11
li $a0, ','
syscall
#updating loop variables
addi $s3, $s3, 1
addi $s7, $s7, 4
j goop



end:
li $v0, 10
syscall
```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Data
Text
Int Regs [10]
PC = 4194640
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 13
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 13
R9 [t1] = 8
R10 [t2] = 0
R11 [t3] = 268697628
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 8
R17 [s1] = 13
R18 [s2] = 21
R19 [s3] = 7
R20 [s4] = 0
R21 [s5] = 0
Text
[004000d8] 000b2021 addu $4, $0, $11 ; 70: move $a0, $t3
Console
Please enter how many fibonacci numbers you want:8
1 number of fibonacci series is stored at memory location 268697600
2 number of fibonacci series is stored at memory location 268697604
3 number of fibonacci series is stored at memory location 268697608
4 number of fibonacci series is stored at memory location 268697612
5 number of fibonacci series is stored at memory location 268697616
6 number of fibonacci series is stored at memory location 268697620
7 number of fibonacci series is stored at memory location 268697624
8 number of fibonacci series is stored at memory location 268697628
fibonacci series: 0,1,1,2,3,5,8,13
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QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194640
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 0
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 1
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 1
R17 [s1] = 0
R18 [s2] = 1
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[004000d8] 000b2021 addu $4, $0, $11 ; 70: move $a0, $t3
Console
Please enter how many fibonacci numbers you want:1
1 number of fibonacci series is stored at memory location 268697600
fibonacci series: 0
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Input Switch Notification
Input language switching
Typing Left Alt + Shift changes your input language. You can turn this feature off or change your hot key sequence by selecting Customize.
Customize
Dismiss



Error handling for non-positive inputs.

```
#Countdown

.data
ent: .asciiz "Please enter the start of the Countdown:"

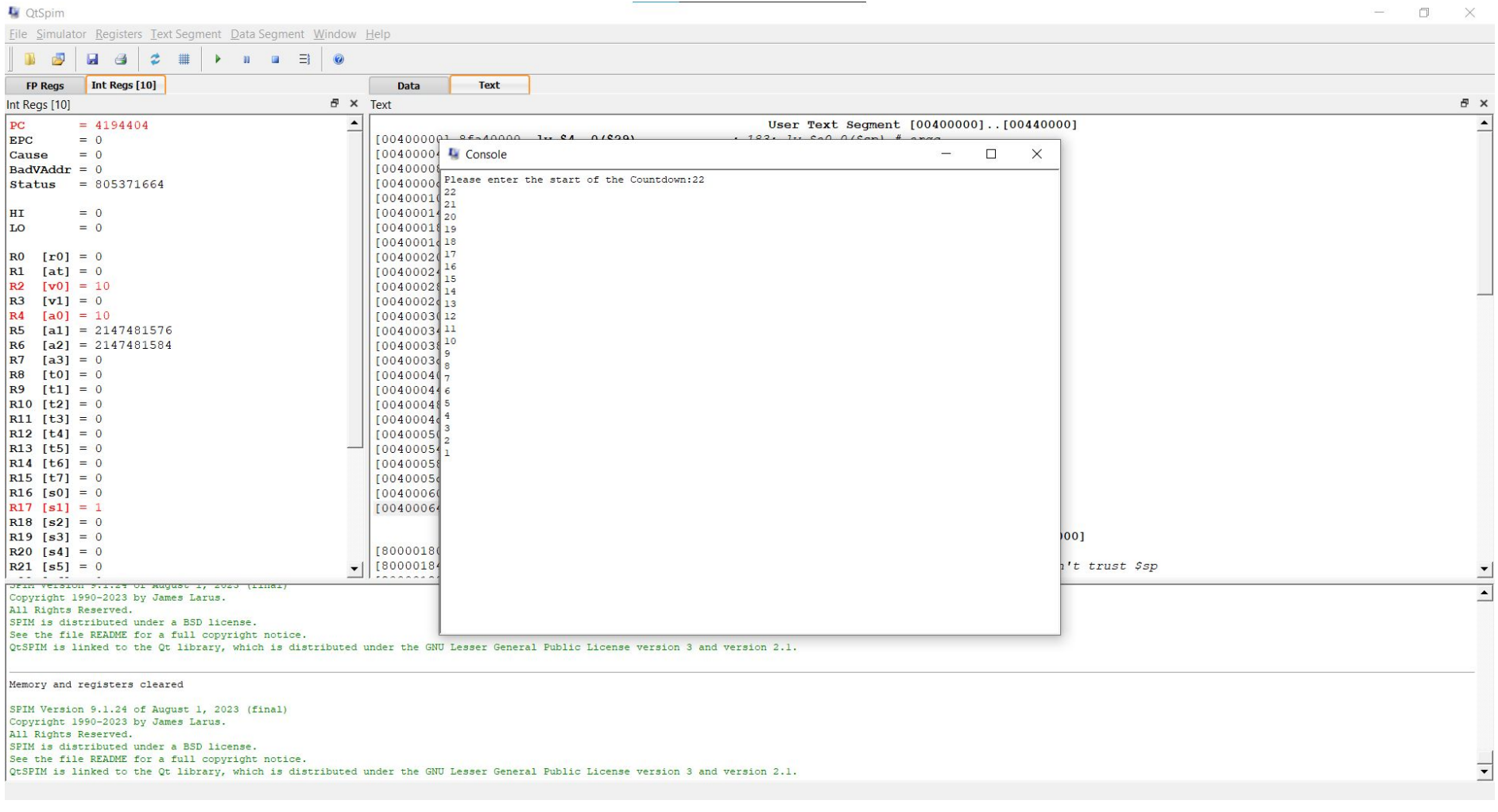
.text
main:
li $v0, 4
la $a0, ent
syscall

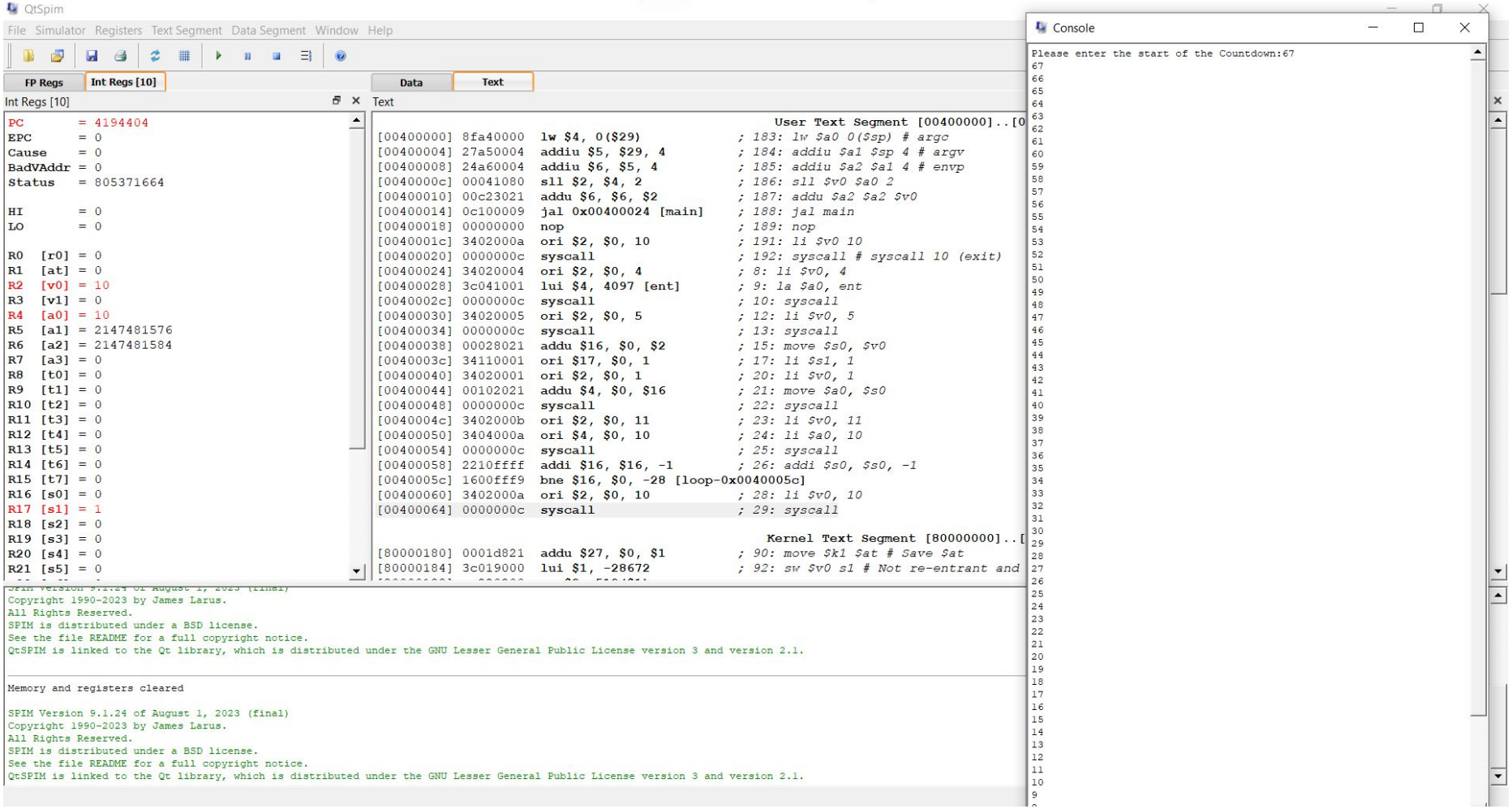
li $v0, 5
syscall

move $s0, $v0

loop:
li $v0, 1
move $a0, $s0
syscall
li $v0, 11
li $a0, 10
syscall
addi $s0, $s0, -1
bne $s0, $0, loop
li $v0, 10
syscall

```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194404
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
User Text Segment [00400000]..[00440000]
Console
Please enter the start of the Countdown:22
22
21
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
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QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194404
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
User Text Segment [00400000]..[0
[00400000] 8fa40000 lw $4, 0($29) ; 183: lw $a0 0($sp) # argc
[00400004] 27a50004 addiu $5, $29, 4 ; 184: addiu $a1 $sp 4 # argv
[00400008] 24a60004 addiu $6, $5, 4 ; 185: addiu $a2 $a1 4 # envp
[0040000c] 00041080 sll $2, $4, 2 ; 186: sll $v0 $a0 2
[00400010] 00c23021 addu $6, $6, $2 ; 187: addu $a2 $a2 $v0
[00400014] 0c100009 jal 0x00400024 [main] ; 188: jal main
[00400018] 00000000 nop ; 189: nop
[0040001c] 3402000a ori $2, $0, 10 ; 191: li $v0 10
[00400020] 0000000c syscall ; 192: syscall # syscall 10 (exit)
[00400024] 34020004 ori $2, $0, 4 ; 8: li $v0, 4
[00400028] 3c041001 lui $4, 4097 [ent] ; 9: la $a0, ent
[0040002c] 0000000c syscall ; 10: syscall
[00400030] 34020005 ori $2, $0, 5 ; 12: li $v0, 5
[00400034] 0000000c syscall ; 13: syscall
[00400038] 00028021 addu $16, $0, $2 ; 15: move $s0, $v0
[0040003c] 34110001 ori $17, $0, 1 ; 17: li $s1, 1
[00400040] 34020001 ori $2, $0, 1 ; 20: li $v0, 1
[00400044] 00102021 addu $4, $0, $16 ; 21: move $a0, $s0
[00400048] 0000000c syscall ; 22: syscall
[0040004c] 3402000b ori $2, $0, 11 ; 23: li $v0, 11
[00400050] 3404000a ori $4, $0, 10 ; 24: li $a0, 10
[00400054] 0000000c syscall ; 25: syscall
[00400058] 2210ffff addi $16, $16, -1 ; 26: addi $s0, $s0, -1
[0040005c] 1600fff9 bne $16, $0, -28 [loop-0x0040005c]
[00400060] 3402000a ori $2, $0, 10 ; 28: li $v0, 10
[00400064] 0000000c syscall ; 29: syscall
Kernel Text Segment [80000000]..[
[80000180] 0001d821 addu $27, $0, $1 ; 90: move $k1 $at # Save $at
[80000184] 3c019000 lui $1, -28672 ; 92: sw $v0 s1 # Not re-entrant and
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Console
Please enter the start of the Countdown:67
67
66
65
64
63
62
61
60
59
58
57
56
55
54
53
52
51
50
49
48
47
46
45
44
43
42
41
40
39
38
37
36
35
34
33
32
31
30
29
28
27
26
25
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10
9

```
#C code copy

.text
main:
li $s0, 1 #J
li $s1, 0 #a

li $t0, 20

while:
bgt $s1, $t0, dowhile
addi $s0, $s0, 1
addi $s1, $s1, 2
j while

dowhile:
li $t0, 100
addi $s0, $s0, 1
bge $s0, $t0, subi
j dowhile

subi:
ble $s1, $0, end
addi $s0, $s0, -1
addi $s1, $s1, -1
j subi

end:
li $v0, 10
syscall


```

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

FP Regs | Int Regs [10] | Data | Text

Int Regs [10]

```
PC          = 4194416
EPC         = 0
Cause       = 0
BadVAddr    = 0
Status      = 805371664

HI          = 0
LO          = 0

R0  [r0]    = 0
R1  [at]    = 0
R2  [v0]    = 10
R3  [v1]    = 0
R4  [a0]    = 1
R5  [a1]    = 2147481576
R6  [a2]    = 2147481584
R7  [a3]    = 0
R8  [t0]    = 100
R9  [t1]    = 0
R10 [t2]    = 0
R11 [t3]    = 0
R12 [t4]    = 0
R13 [t5]    = 0
R14 [t6]    = 0
R15 [t7]    = 0
R16 [s0]    = 78
R17 [s1]    = 0
R18 [s2]    = 0
R19 [s3]    = 0
R20 [s4]    = 0
R21 [s5]    = 0
```

Text

```
                    User Text Segment [00400000]..[00440000]
[00400000] 8fa40000  lw $4, 0($29)            ; 183: lw $a0 0($sp) # argc
[00400004] 27a50004  addiu $5, $29, 4         ; 184: addiu $a1 $sp 4 # argv
[00400008] 24a60004  addiu $6, $5, 4          ; 185: addiu $a2 $a1 4 # envp
[0040000c] 00041080  sll $2, $4, 2            ; 186: sll $v0 $a0 2
[00400010] 00c23021  addu $6, $6, $2          ; 187: addu $a2 $a2 $v0
[00400014] 0c100009  jal 0x00400024 [main]    ; 188: jal main
[00400018] 00000000  nop                      ; 189: nop
[0040001c] 3402000a  ori $2, $0, 10           ; 191: li $v0 10
[00400020] 0000000c  syscall                  ; 192: syscall # syscall 10 (exit)
[00400024] 34100001  ori $16, $0, 1           ; 7: li $s0, 1 #J
[00400028] 34110000  ori $17, $0, 0           ; 8: li $s1, 0 #a
[0040002c] 34080014  ori $8, $0, 20           ; 10: li $t0, 20
[00400030] 0111082a  slt $1, $8, $17          ; 13: bgt $s1, $t0, dowhile
[00400034] 14200004  bne $1, $0, 16 [dowhile-0x00400034]
[00400038] 22100001  addi $16, $16, 1         ; 14: addi $s0, $s0, 1
[0040003c] 22310002  addi $17, $17, 2         ; 15: addi $s1, $s1, 2
[00400040] 0810000c  j 0x00400030 [while]     ; 16: j while
[00400044] 34080064  ori $8, $0, 100          ; 19: li $t0, 100
[00400048] 22100001  addi $16, $16, 1         ; 20: addi $s0, $s0, 1
[0040004c] 0208082a  slt $1, $16, $8          ; 21: bge $s0, $t0, subi
[00400050] 10200002  beq $1, $0, 8 [subi-0x00400050]
[00400054] 08100011  j 0x00400044 [dowhile]   ; 22: j dowhile
[00400058] 0011082a  slt $1, $0, $17          ; 25: ble $s1, $0, end
[0040005c] 10200004  beq $1, $0, 16 [end-0x0040005c]
[00400060] 2210ffff  addi $16, $16, -1        ; 26: addi $s0, $s0, -1
[00400064] 2231ffff  addi $17, $17, -1        ; 27: addi $s1, $s1, -1
[00400068] 08100016  j 0x00400058 [subi]      ; 28: j subi
[0040006c] 3402000a  ori $2, $0, 10           ; 31: li $v0, 10
[00400070] 0000000c  syscall                  ; 32: syscall
```

Final values in s0 (J) and s1 (a).

```
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spim: (parser) syntax error on line 28 of file C:/Users/valorant_guy/Desktop/Computer)rganizatin/assignment4/a4q3.S
  sub
  ^

Memory and registers cleared

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```

Divyam Sood 23112

```
#Triangle

.data
ent: .asciiz "Please enter the k:"

.text
main:
li $v0, 4
la $a0, ent
syscall
li $v0, 5
syscall

move $s0, $v0 #k
addi $s1, $s0, -1 #k-1
li $s7, 0 #counter of how many lines, will go till k

#3 loops: one to print spaces, one to print ascending number, one to print descending number. whole thing wrapped in a bigger loop.

big:
bge $s7, $s0, exit
sub $t0, $s0, $s1 #the maximum number (numerically) in the line

li $t1, 1
spaces:
beq $s1, $0, after_spaces
li $v0, 11
li $a0, ' '
syscall
addi $t1, $t1, 1
ble $t1, $s1, spaces

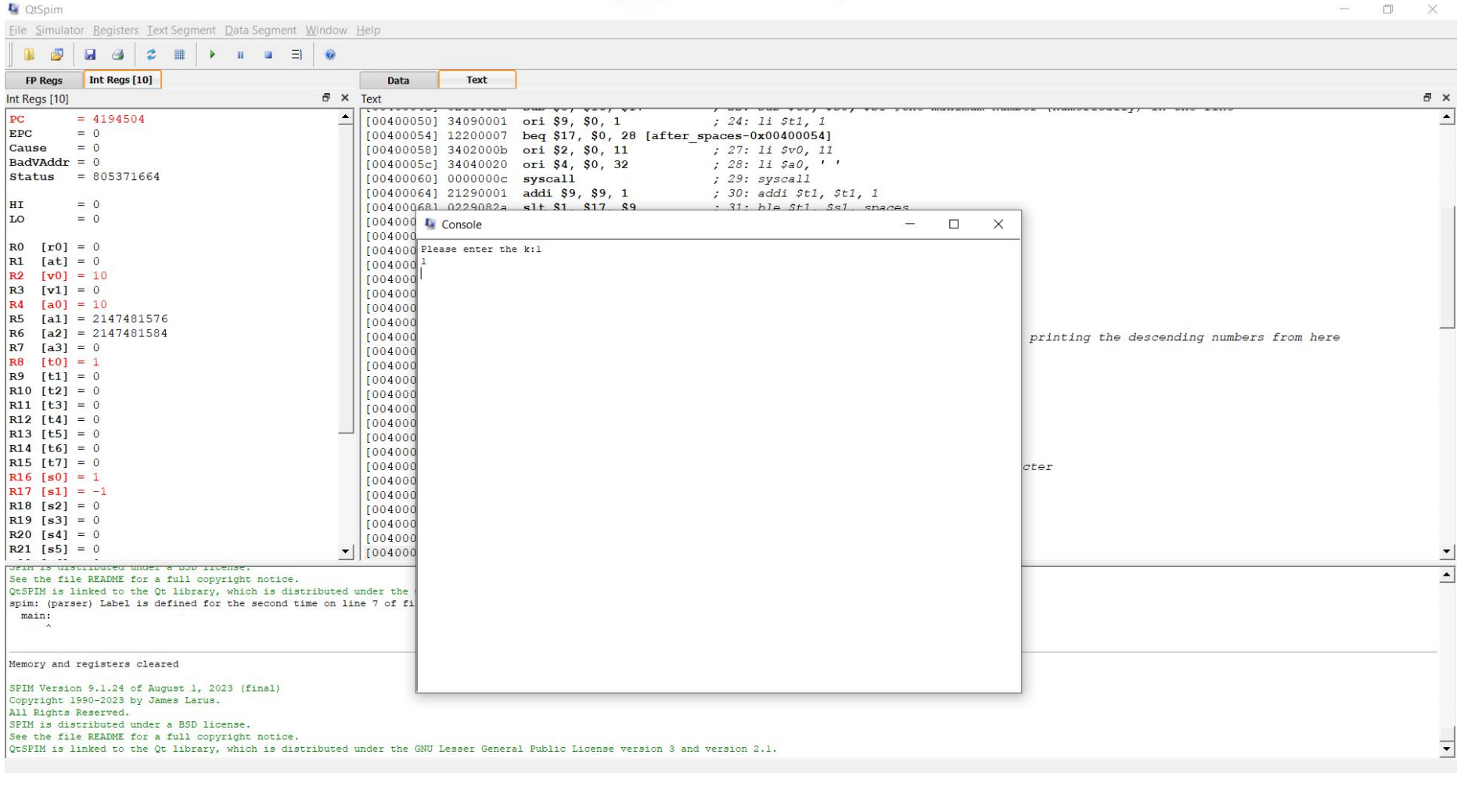

after_spaces:
li $t1, 1
asc:
li $v0, 1
move $a0, $t1
syscall
addi $t1, $t1, 1
ble $t1, $t0, asc

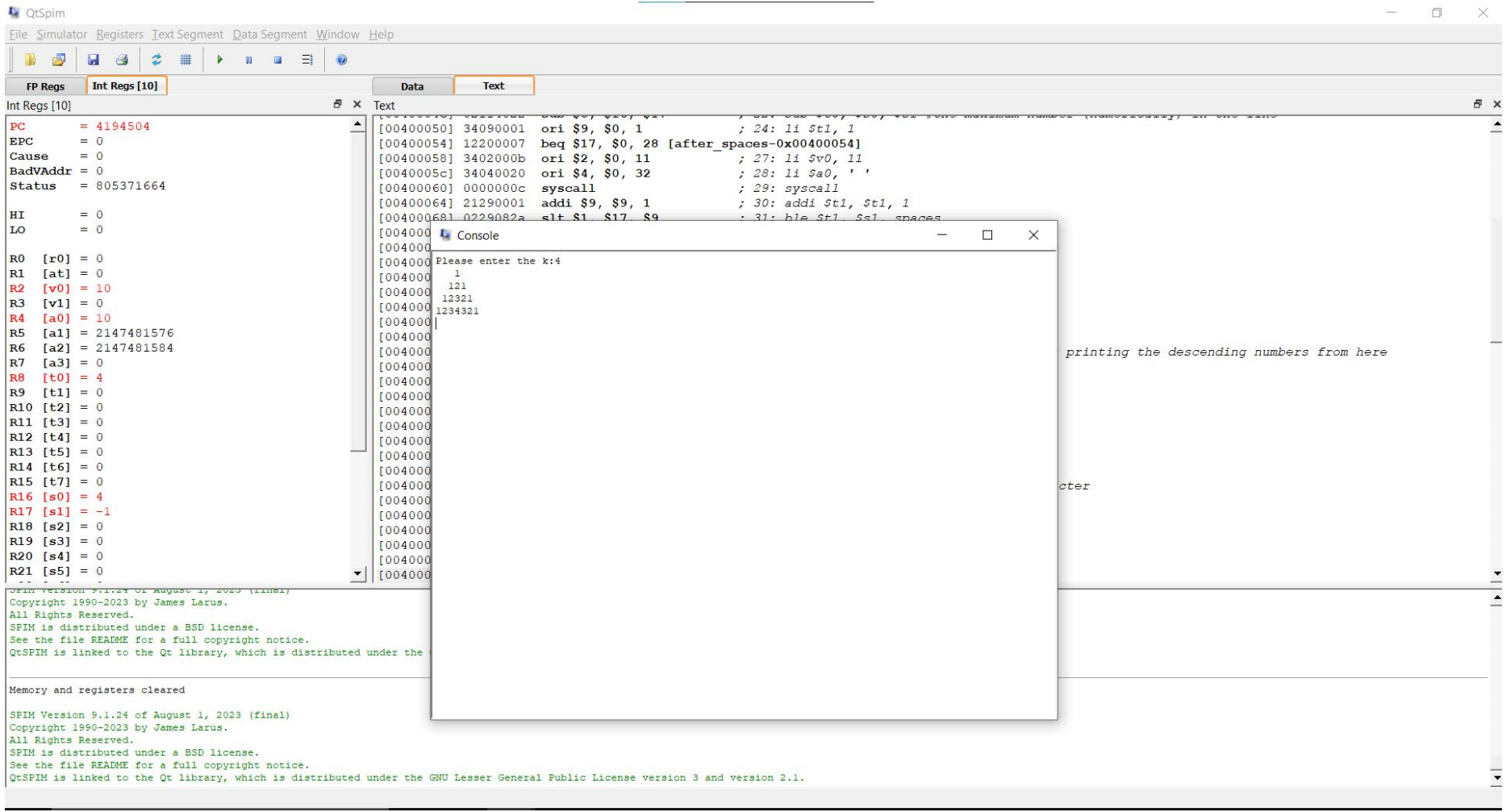
addi $t1, $t1, -2 #we have to start printing the descending numbers from here
dsc:
ble $t1, $0, blu
li $v0, 1
move $a0, $t1
syscall
addi $t1 $t1, -1
j dsc


blu:
#big loop updation (blu)
li $v0, 11
li $a0, 10 #ASCII for newline character
syscall
addi $s7, $s7, 1
addi $s1, $s1, -1
j big

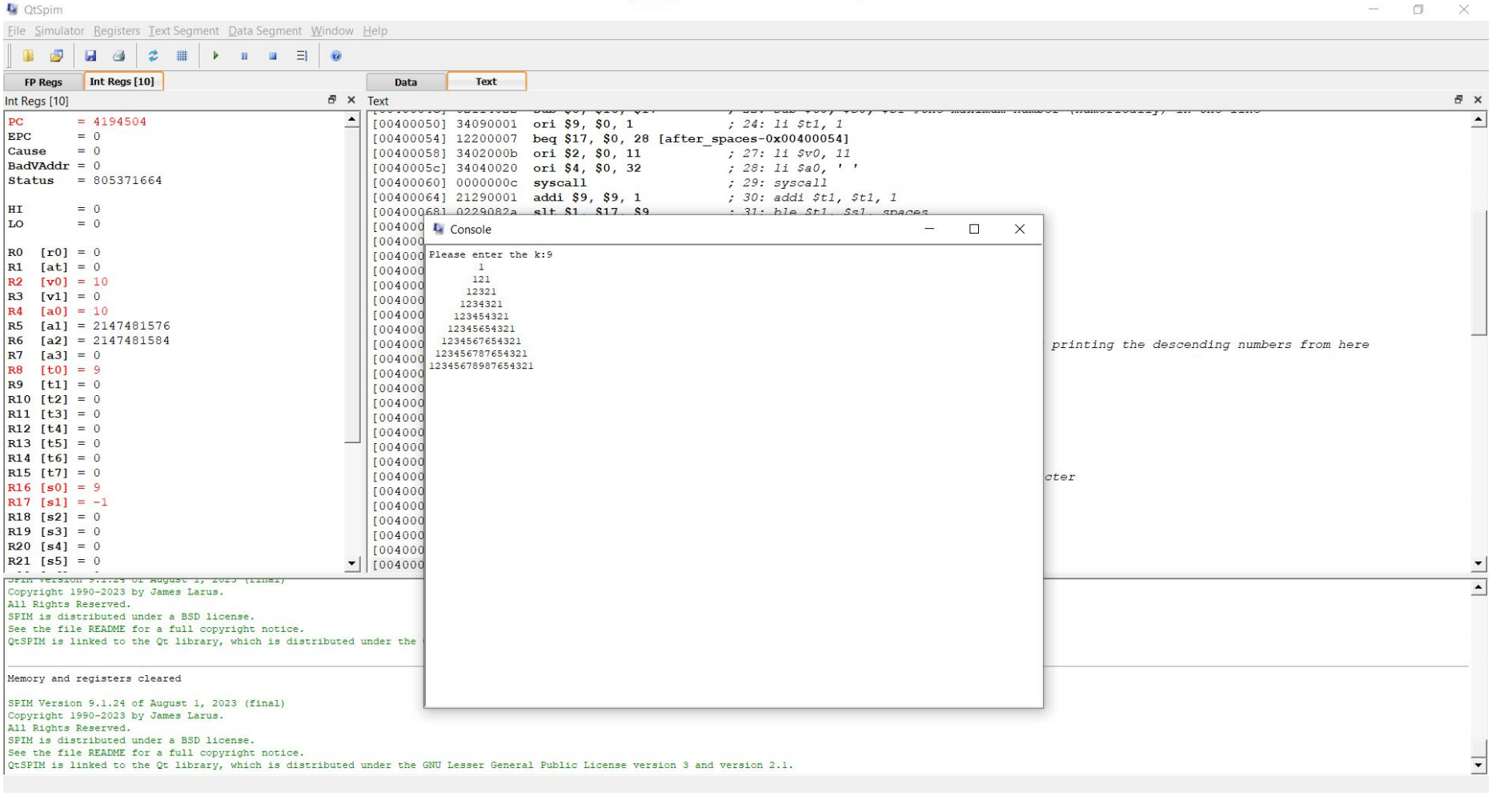

exit:
li $v0, 10
syscall
```

Divyam Sood 23112


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194504
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 1
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 1
R17 [s1] = -1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[00400050] 34090001 ori $9, $0, 1 ; 24: li $t1, 1
[00400054] 12200007 beq $17, $0, 28 [after_spaces-0x00400054]
[00400058] 3402000b ori $2, $0, 11 ; 27: li $v0, 11
[0040005c] 34040020 ori $4, $0, 32 ; 28: li $a0, ' '
[00400060] 0000000c syscall ; 29: syscall
[00400064] 21290001 addi $9, $9, 1 ; 30: addi $t1, $t1, 1
printing the descending numbers from here
Console
Please enter the k:1
1
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spim: (parser) Label is defined for the second time on line 7 of fi
main:
^
Memory and registers cleared
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Divyam Sood 23112


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194504
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 0
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 10
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 9
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 9
R17 [s1] = -1
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[00400050] 34090001 ori $9, $0, 1 ; 24: li $t1, 1
[00400054] 12200007 beq $17, $0, 28 [after_spaces-0x00400054]
[00400058] 3402000b ori $2, $0, 11 ; 27: li $v0, 11
[0040005c] 34040020 ori $4, $0, 32 ; 28: li $a0, ' '
[00400060] 0000000c syscall ; 29: syscall
[00400064] 21290001 addi $9, $9, 1 ; 30: addi $t1, $t1, 1
printing the descending numbers from here
cter
Console
Please enter the k:9
1
121
12321
1234321
123454321
12345654321
1234567654321
123456787654321
12345678987654321
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